

Wooseong Jeong

Ph.D candidate in KAIST
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PERSONAL DATA

- **Birth / Nationality:** 27th April, 1997 / Republic of Korea
- **Language:** First language Korean, Fluent in English

EDUCATION

- **Korea Advance Institute of Science and Technology (KAIST)** Daejeon, Korea
• Ph.D candidate in Mechanical Engineering 03/2023 – 08/2027 (Expected)
Advisor: Kuk-Jin Yoon
- **Korea Advance Institute of Science and Technology (KAIST)** Daejeon, Korea
• MS in Mechanical Engineering March 2021 - February 2023
Advisor: Kuk-Jin Yoon
Thesis: Connection Strength-based Optimization with Progressive Multi-modal Feature Exchange for Multi-task Learning
- **Korea Advance Institute of Science and Technology (KAIST)** Daejeon, Korea
• Bachelor of Engineering March 2015 - February 2021
Double Major: Mechanical Engineering and Electrical Engineering, cum laude
Dean's List: Fall, 2017
- **Daegu Il Science High School** Daegu, Korea
• Early graduation March 2013 - February 2015

RESEARCH INTERESTS

- **Foundation Model Optimization and Adaptation**
 - Multi-task, multi-objective, and non-convex optimization
 - Parameter-efficient adaptation, model merging, and knowledge distillation (including on-policy distillation)
 - Test-time adaptation and in-context learning
- **Embodied Intelligence**
 - Vision-language-action models and uncertainty-guided reasoning
 - Robust perception and motion planning for autonomous systems

PUBLICATIONS

- **Wooseong Jeong** and Kuk-Jin Yoon, “The Coupling Is the Lever: Cross-Group Geometry for Joint Optimization”, International Conference on Learning Representations (ICLR) 2027 (Under Review).
- Giwon Lee, **Wooseong Jeong**, and Kuk-Jin Yoon, “Context Distillation Distills More Than Skills: Navigating Skill Gains and Cross-Domain Retention with State-Shared Drift”, International Conference on Learning Representations (ICLR) 2027 (Under Review).
- Giwon Lee*, **Wooseong Jeong***, and Kuk-Jin Yoon, “Beyond Context Selection: Consensus over Responses to Alternative Contexts”, International Conference on Learning Representations (ICLR) 2027 (Under Review).
- **Wooseong Jeong***, Yujeong Chae*, and Kuk-Jin Yoon, “LiRA-Merge: Online Model Merging for Test-Time Modality Rebalancing in LiDAR–RADAR 3D Detection”, AAAI Conference on Artificial Intelligence (AAAI) 2027 (Under Review).
- **Wooseong Jeong***, Giwon Lee*, and Kuk-Jin Yoon, “Latent Reasoning Path Guidance: Adapting ICL Samplers for Cross-Domain Retrieval”, Annual Meeting of the Association for Computational Linguistics (ACL) (Under Review).
- **Wooseong Jeong**, Giwon Lee, and Kuk-Jin Yoon, “CARMA: Cross-Domain Module Merging for Robust Motion Planning and End-to-End Driving”, IEEE Transactions on Intelligent Vehicles (T-IV) (Under Review).
- Hyemin Yang*, **Wooseong Jeong***, Giwon Lee*, and Kuk-Jin Yoon, “Triggering Generalist Reasoning via Predictive Uncertainty for Dual-System VLA”, Conference on Neural Information Processing Systems (NeurIPS) 2026 (Accepted).

- **Wooseong Jeong***, Wonyoung Lee*, and Kuk-Jin Yoon, "Preference-Aligned LoRA Merging: Preserving Subspace Coverage and Addressing Directional Anisotropy", IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- Wonyoung Lee*, **Wooseong Jeong*** and Kuk-Jin Yoon, "Label-Free Cross-Task LoRA Merging with Null-Space Compression", IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- **Wooseong Jeong*** Jegyeong Cho*, Youngho Yoon*, Jaeyoung Lee and Kuk-Jin Yoon, "Stabilizing Multi-Task Latent Spaces: Recursive Refinement with Coordinators in Partially Labeled Learning", IEEE Access 2026.
- Giwon Lee*, **Wooseong Jeong***, Daehee Park, Jaewoo Jeong, and Kuk-Jin Yoon, "Interaction-Merged Motion Planning: Effectively Leveraging Diverse Motion Datasets for Robust Planning", The International Conference on Computer Vision (ICCV) 2025. **(Highlight)**
- **Wooseong Jeong***, and Kuk-Jin Yoon, "Resolving Token-Space Gradient Conflicts: Token Space Manipulation for Transformer-Based Multi-Task Learning", The International Conference on Computer Vision (ICCV) 2025.
- **Wooseong Jeong***, Jegyeong Cho*, Youngho Yoon*, and Kuk-Jin Yoon, "Synchronizing Task Behavior: Aligning Multiple Tasks during Test-Time Training", The International Conference on Computer Vision (ICCV) 2025.
- Jihun Kim*, Hoyong Kwon*, Hyeokjun Kwon*, **Wooseong Jeong**, and Kuk-Jin Yoon, "DC-TTA: Divide-and-Conquer Framework for Test-Time Adaptation of Interactive Segmentation", The International Conference on Computer Vision (ICCV) 2025.
- **Wooseong Jeong***, Jihun Kim*, Hyeokjun Kwon*, and Kuk-Jin Yoon, "Multi-View 3D Scene Abstraction From Drone-Captured RGB Images", IEEE Access 2025.
- **Wooseong Jeong** and Kuk-Jin Yoon, "Selective Task Group Updates for Multi-Task Optimization", The International Conference on Learning Representations (ICLR) 2025.
- **Wooseong Jeong** and Kuk-Jin Yoon, "Quantifying Task Priority for Multi-Task Optimization", IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
- Hyeokjun Kwon*, Hyeonseong Kim*, Yoonsu Kang*, Youngho Yoon*, **Wooseong Jeong**, and Kuk-Jin Yoon, "Pixel-wise Warping for Deep Image Stitching" Thirty-Seventh AAAI Conference on Artificial Intelligence (AAAI) 2023

PATENTS

- Jae Young Lee, Seung Uk Ahn, Young Woo Seo, Kuk-Jin Yoon, **Wooseong Jeong**, Je Gyeong Cho, and Young Ho Yoon, "Apparatus and Method for Learning Multiple Tasks in a Partial Labeling Environment", Republic of Korea Patent Application No. 10-2026-0089936, Filed May 18, 2026.
- Kuk-Jin Yoon, Hyeokjun Kwon, Yoonsu Kang, **Wooseong Jeong**, Hyeonseong Kim, and Hwi-Sung Park, "Apparatus and Method for Generating Image Dataset for Learning and Evaluation of Image Stitching", Republic of Korea Patent No. 10-2645640, Registered Mar. 5, 2024 (Application No. 10-2021-0157112, Filed Nov. 15, 2021).
- Kuk-Jin Yoon, Hyeokjun Kwon, Yoonsu Kang, **Wooseong Jeong**, Hyeonseong Kim, and Hwi-Sung Park, "Apparatus and Method for Image Stitching", Republic of Korea Patent No. 10-2571530, Registered Aug. 23, 2023 (Application No. 10-2021-0157069, Filed Nov. 15, 2021).

RESEARCH EXPERIENCES

- **Project Leader** March 2025 - Present
 - Core Technology Development for Autonomous Driving in Unstructured Off-Road Environments
 - Funded by Hanwha Aerospace
- **Project Leader** March 2024 - March 2025
 - Unmanned Swarm CPS Research Lab
 - Funded by Agency for Defense Development, ADD
- **Project Leader** March 2022 - February 2024
 - Development of a 2kg-Class Small EOTS (Electro-Optical Tracking System) for Drones
 - Funded by Institute of Civil-Military Technology Cooperation
- **Project Member** March 2021 - February 2022
 - Unmanned Swarm CPS Research Lab
 - Funded by Agency for Defense Development, ADD

ACADEMIC REVIEWER

- **AAAI:** (2027)
- **NeurIPS:** (2024, 2025)
- **ICLR:** (2025, 2026)
- **CVPR:** (2025, 2026)
- **ICML:** (2025)
- **ICCV:** (2025)
- **ECCV:** (2026)

SKILLS SUMMARY

- **Languages:** Python (proficient), C++ (proficient), MATLAB
- **Frameworks & Libraries:** PyTorch, JAX, NumPy, OpenCV
- **Tools:** ROS, Git, Linux